

ITLIA0-44-Formative Project-Mowbray Campus-5737241

Question 1 - VirtualBox with Bridged Adapter

Section 1: Virtualization Setup

1. Install VirtualBox

Download the latest stable version of VirtualBox from the official Oracle website.



2. Create and Configure Ubuntu VM

Click New in VirtualBox Manager.

Name: Ubuntu_alpha

Type: Linux.

Version: Ubuntu (64-bit).

Memory (RAM): Allocate 4096 MB (4 GB)

Hard Disk: Allocate at least 20 GB.

General

Name: Ubuntu_alpha

Operating System: Ubuntu (64-bit)

System

Base Memory: 4048 MB

Boot Order: Floppy, Optical, Hard Disk

Acceleration: Nested Paging, KVM Paravirtualization

Display

Video Memory: 16 MB

Graphics Controller: VMSVGA

Remote Desktop Server: Disabled

Recording: Disabled

Storage

Controller: IDE

IDE Secondary Device 0: [Optical Drive] ubuntu-24.04.2-desktop-amd64.iso (5,91 GB)

Controller: SATA

SATA Port 0: Ubuntu_alpha.vdi (Normal, 25.00 GB)

Network Settings:

Go to Network.

Adapter 1: Select Bridged Adapter.

Network

Adapter 1 Adapter 2 Adapter 3 Adapter 4

☒ Enable Network Adapter

Attached to: Bridged Adapter

Name: Intel(R) Wi-Fi 6 AX203

Adapter Type: Intel PRO/1000 MT Desktop (82540EM)

Promiscuous Mode: Deny

MAC Address: 080027EA320D

☒ Cable Connected

Section 2: Ubuntu OS Installation

3. Install Ubuntu OS

Mount the Ubuntu ISO image and start the VM.

Select the desired language and continue.

Select your timezone

Location: Cape Town (Western Cape, South Africa)Timezone: Africa/Johannesburg



BackProgress indicator: 10 dots, 10th dot highlightedNext

Select the correct keyboard layout.

Keyboard layout



Select your keyboard layoutDetect

English (UK)
English (US)
Esperanto

Select your keyboard variant:English (UK)

Type here to test your keyboard

BackProgress indicator: 10 dots, 2nd dot highlightedNext

Choose Minimal installation

Applications



What apps would you like to install to start with?

☒ **Default selection**
Just the essentials, web browser and basic utilities.

☐ **Extended selection**
An offline-friendly selection of office tools, utilities and web browser.

Back

Progress indicator: 10 dots, 4th dot highlighted

Next

Personal Details:

Use your own name.

Choose a username and password that conforms to Linux security standards

Create your account

Create your account

Your name
ameer

Your computer's name
ameer-VirtualBox

Your username
ameer

Password
Password123

Confirm password
Password123

Hide

Weak password

☐ Require my password to log in

☐ Use Active Directory

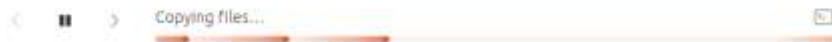
Progress indicator: 10 dots, 4th dot highlighted

Next

Install the OS

Fast, free
and full of
new
features

The latest version makes
computing easier than ever.
Whether you're a developer,
creator, gamer or administrator
you'll find new tools to improve
your productivity and enhance



Post-Installation Steps:

The assignment requires the hostname to be alpha. This is typically set during installation or can be modified post-installation using `hostnamectl`.

Installation complete



Ubuntu 24.04.2 LTS is installed and ready to use

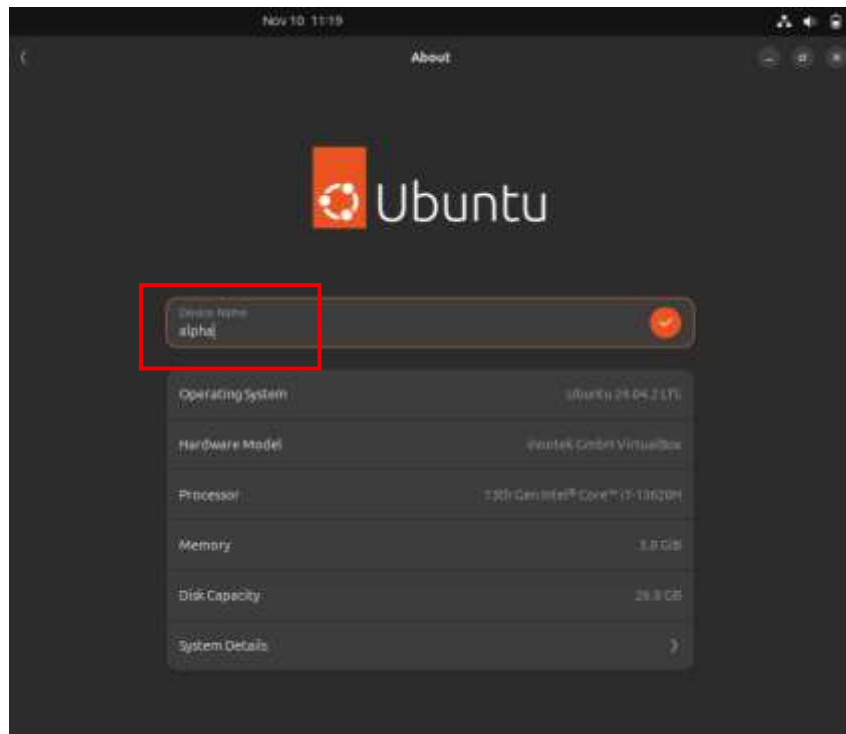
Restart to complete the installation or continue testing.
Any changes you make will not be saved.

Continue testing

Restart now

Restart the VM after the installation is complete.

Time and Date Setting:

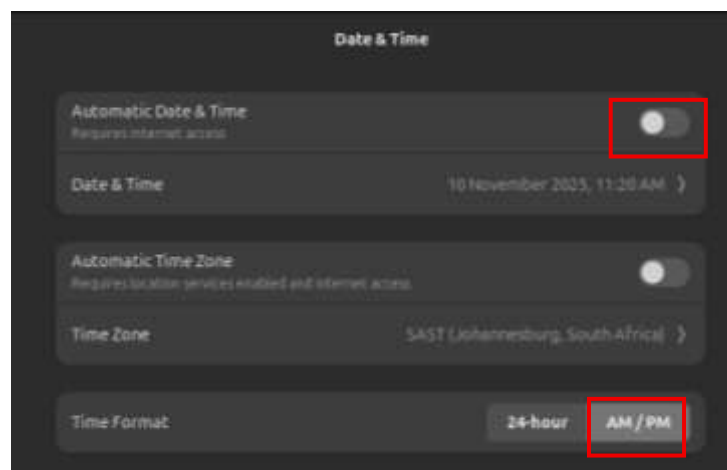


Navigate to Settings

Find Date & Time or Clock.

Disable automatic time synchronization. Set the time manually.

Adjust the clock format setting to show time in 12-hour format (AM/PM).



Section 3: Configure Networking in Ubuntu

4. Configure Networking

Enable Network Card (if down) and Assign IP:

“sudo ip link set dev <Interface_Name> up”

```
aneer@aneer-VirtualBox: ~/Desktop$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:53:b3:2d brd ff:ff:ff:ff:ff:ff
    inet 192.168.68.51/22 brd 192.168.71.255 scope global dynamic noprefixroute enp0s3
        valid_lft 6736sec preferred_lft 6736sec
    inet6 fe80::a00:27ff:fe53:b32d/64 scope link
        valid_lft forever preferred_lft forever
aneer@aneer-VirtualBox: ~/Desktop$
```

“sudo ip addr add <Linux_IP_Address>/24 dev <Interface_Name>”

```
aneer@aneer-VirtualBox: ~/Desktop$ sudo ip addr add 192.168.68.51/22 dev enp0s3
aneer@aneer-VirtualBox: ~/Desktop$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:53:b3:2d brd ff:ff:ff:ff:ff:ff
    inet 192.168.68.51/22 scope global enp0s3
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe53:b32d/64 scope link
        valid_lft forever preferred_lft forever
aneer@aneer-VirtualBox: ~/Desktop$
```

Netplan files are typically in `/etc/netplan/`. Look for a file like `00-installer-config.yaml` or similar.

“ls /etc/netplan/”

Use a professional text editor like nano or vim.

“sudo nano /etc/netplan/00-installer-config.yaml”

Modify the file to look similar to the structure below. Indentation is crucial in YAML.

Replace `<Linux_IP_Address>` with your chosen static IP (10.1.1.20).

The subnet mask is represented by `/24` (equivalent to 255.255.255.0).

“network:

version: 2

renderer: networkd

ethernets:

<Interface_Name>: # e.g., enp0s3

dhcp4: no # Disable DHCP

addresses: [192.168.68.51/22] # Static IP and Subnet (This allows the ping)

routes:

- to: default

via: 10.1.1.101 # Gateway required by assignment, even though it won't work on this network

nameservers:

addresses: [10.1.1.110] # DNS Server required by assignment"

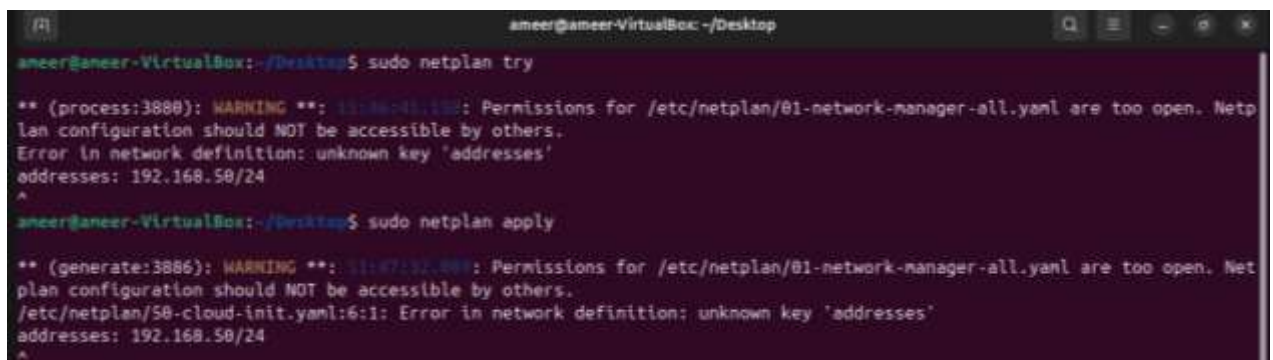


```
GNU nano 7.2 /etc/netplan/50-cloud-init.yaml
network:
  version: 2
  renderer: networkd
  ethernet3:
    enp4s3:
      dhcp4: no
      addresses: 192.168.68.51/22
      routes:
        - to: default
          via: 10.1.1.101
      nameservers:
        addresses: 10.1.1.110
```

Restart Networking Service

"sudo netplan try # Optional: Test configuration before applying

sudo netplan apply # Apply the changes persistently"



```
ameer@ameer-VirtualBox: ~/Desktop
ameer@ameer-VirtualBox:~/Desktop$ sudo netplan try
** (process:3880): WARNING **: 11:06:41.180: Permissions for /etc/netplan/01-network-manager-all.yaml are too open. Netplan configuration should NOT be accessible by others.
Error in network definition: unknown key 'addresses'
addresses: 192.168.50/24
^
ameer@ameer-VirtualBox:~/Desktop$ sudo netplan apply
** (generate:3886): WARNING **: 11:07:32.899: Permissions for /etc/netplan/01-network-manager-all.yaml are too open. Netplan configuration should NOT be accessible by others.
/etc/netplan/50-cloud-init.yaml:6:1: Error in network definition: unknown key 'addresses'
addresses: 192.168.50/24
^
```

Edit the Hosts File:

"sudo nano /etc/hosts"

Add entries for the Linux machine and the Windows 10 machine. Replace the placeholder IP addresses.

"# --- Start of Assignment Entries ---"

<Linux_IP_Address> alpha

<Windows_10_IP> win10host

--- End of Assignment Entries ---"

```
GNU nano 7.2 /etc/hosts
192.168.68.50 localhost
192.168.68.55 alpha

# The following lines are desirable for IPv6 capable hosts
::1 ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
```

Adding entries to /etc/hosts ensures that the systems can resolve each other's hostnames immediately without relying on the DNS server (10.1.1.110), offering a layer of redundancy and faster local resolution.

Section 4: Verify Connectivity

5. Verify Connectivity

Use the ping command with the count flag (-c) to send exactly three requests to the Windows 10 machine.

"ping -c 4 <Windows_10_IP>"

```
ameer@ameer-VirtualBox:~/Desktop$ ping -c 3 192.168.68.50
PING 192.168.68.50 (192.168.68.50) 56(84) bytes of data:
... 192.168.68.50 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2055ms
ameer@ameer-VirtualBox:~/Desktop$
```

Successful output will show three "3 received" packets and a "0% packet loss" summary.

If the ping fails, troubleshoot in this order:

- Check the Linux IP (ip a). Is it the correct static address?
- Check the Netplan configuration. Did you run `sudo netplan apply`?
- Check the Windows 10 machine's Firewall settings (most common issue).
- Verify the Bridged Adapter is set correctly in VirtualBox settings and is using the active host network interface

```
ameer@ameer-VirtualBox:~/Desktop$ ping -c 3 192.168.68.50
PING 192.168.68.50 (192.168.68.50) 56(84) bytes of data:
64 bytes from 192.168.68.50: icmp_seq=1 ttl=128 time=0.508 ms
64 bytes from 192.168.68.50: icmp_seq=2 ttl=128 time=0.455 ms
64 bytes from 192.168.68.50: icmp_seq=3 ttl=128 time=1.08 ms

--- 192.168.68.50 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2036ms
rtt min/avg/max/mdev = 0.455/0.682/1.084/0.284 ms
ameer@ameer-VirtualBox:~/Desktop$
```

2.1 Run the necessary command-line tool that will allow the Linux machine to run files that have an .exe extension. Install VLC on the Linux machine.

Install Wine:

“sudo apt update”

```
aneer@aneer-VirtualBox: ~/Desktop$ sudo apt update
[sudo] password for aneer:
Sorry, try again.
[sudo] password for aneer:
Hit:1 http://za.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:3 http://za.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://za.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:5 http://za.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
Get:6 http://za.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:7 http://za.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [378 kB]
Get:8 http://za.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:9 http://za.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7,124 B]
Get:10 http://za.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Get:11 http://za.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [11.0 kB]
Get:12 http://za.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Get:13 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.6 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:15 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [52.1 kB]
Get:16 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Fetched 1,025 kB in 3s (378 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
1 package can be upgraded. Run 'apt list --upgradable' to see it.
aneer@aneer-VirtualBox: ~/Desktop$
```

“sudo apt install wine”

```

amber@ameer-VirtualBox: ~/Desktop$ sudo apt install wine
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libgl1-amd-gpu-dri libgltapi-mesa libllvm9
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  fonts-wine gcc-14-base:i386 glib-networking:i386 gststreamer1.0-plugins-base:i386 gststreamer1.0-plugins-good:i386
  gststreamer1.0-x:i386 i965-vd-driver:i965-vd-driver:i386 intel-media-vd-driver:i386 intel-media-va-driver:i386 libaa1:i386
  libao3:i386 libapparmor1:i386 libasound2-plugins:i386 libasound2-t64:i386 libasyncns0:i386
  libatk-bridge2.0-0t64:i386 libatk1.0-0t64:i386 libatomic1:i386 libatspi2.0-0t64:i386 libavahi-client3:i386
  libavahi-common-data:i386 libavahi-common3:i386 libavc1394-0:i386 libavcodec60:i386 libavcodec68:i386 libavutil58
  libavutil58:i386 libblkid1:i386 libbluetooth3:i386 libbsd0:i386 libbz2-1.0:i386 libc6:i386 libcacado3:i386
  libcairo-gobject2:i386 libcairo2:i386 libcap2:i386 libcap120-3t64 libcap120-3t64:i386 libcdparanoia0:i386

```

“wine-version”

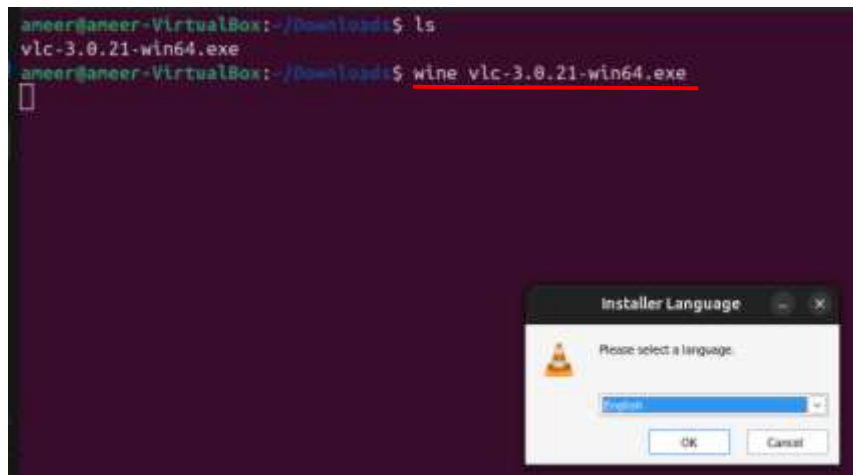
```
Setting up gstreamer1.0-plugins-good:1386 (1.24.2-1ubuntu1.2) ...
Processing triggers for wine (9.0-repack-4build3) ...
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...
Processing triggers for libgdk-pixbuf-2.0-0:1386 (2.42.10+dfsg-3ubuntu3.2) ...
aneer@aneer-VirtualBox: ~/Desktop$ wine --version
wine-9.0 (Ubuntu 9.0-repack-4build3)
aneer@aneer-VirtualBox: ~/Desktop$
```

Install VLC (using the .exe file):

Download the official Windows installer for VLC.

Run the installer using the wine command:

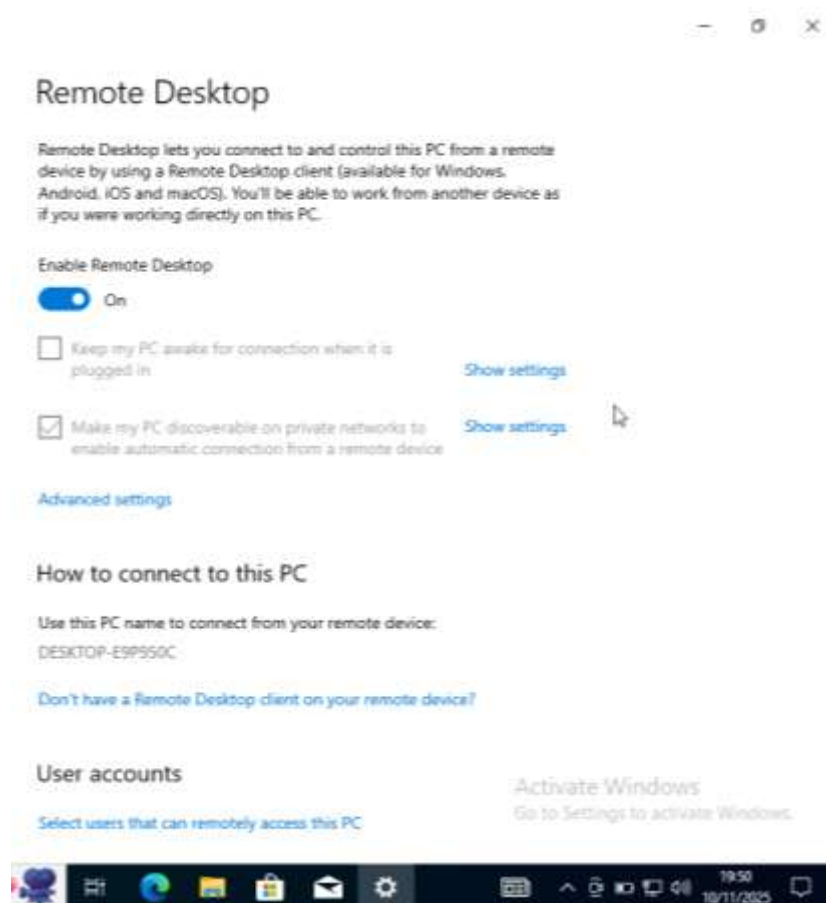
`"wine ~/Downloads/vlc-x.x.x-win32.exe"`



2.2 Configure the Windows 10 machine to allow remote desktop connections.

On the Windows 10 machine, go to Settings > System > Remote Desktop.

Toggle the Enable Remote Desktop setting to On.



2.3 Ensure that you can connect to the Windows machine from the Linux machine using Remmina Remote Desktop Client and map the local hard drive to the remote Windows machine. Verify that the Linux Drive appears on the Windows 10 Machine.

Remmina Setup (Linux):

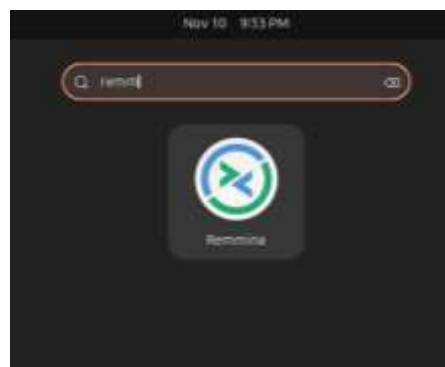
Install Remmina: If not already installed on Ubuntu, install the client:

“sudo apt install remmina remmina-plugin-rdp”

```
aneer@aneer-VirtualBox:~/Downloads$ ls
vlc-3.0.21-win64.exe
aneer@aneer-VirtualBox:~/Downloads$ wine vlc-3.0.21-win64.exe
aneer@aneer-VirtualBox:~/Downloads$ sudo apt install remmina remmina-plugin-rdp
[sudo] password for aneer:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
remmina is already the newest version (1.4.35+dfsg-0ubuntu5.1).
remmina set to manually installed.
remmina-plugin-rdp is already the newest version (1.4.35+dfsg-0ubuntu5.1).
remmina-plugin-rdp set to manually installed.
The following packages were automatically installed and are no longer required:
  libglib2.0-0 libglapi-mesa libllvm19
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 1 not upgraded.
aneer@aneer-VirtualBox:~/Downloads$
```

Create RDP Connection Profile:

Launch Remmina and click the + icon to create a new connection profile.



Protocol: Select RDP - Remote Desktop Protocol.

Server: Enter the Windows 10 IP address (10.0.2.7)

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\ameer> ipconfig

Windows IP Configuration

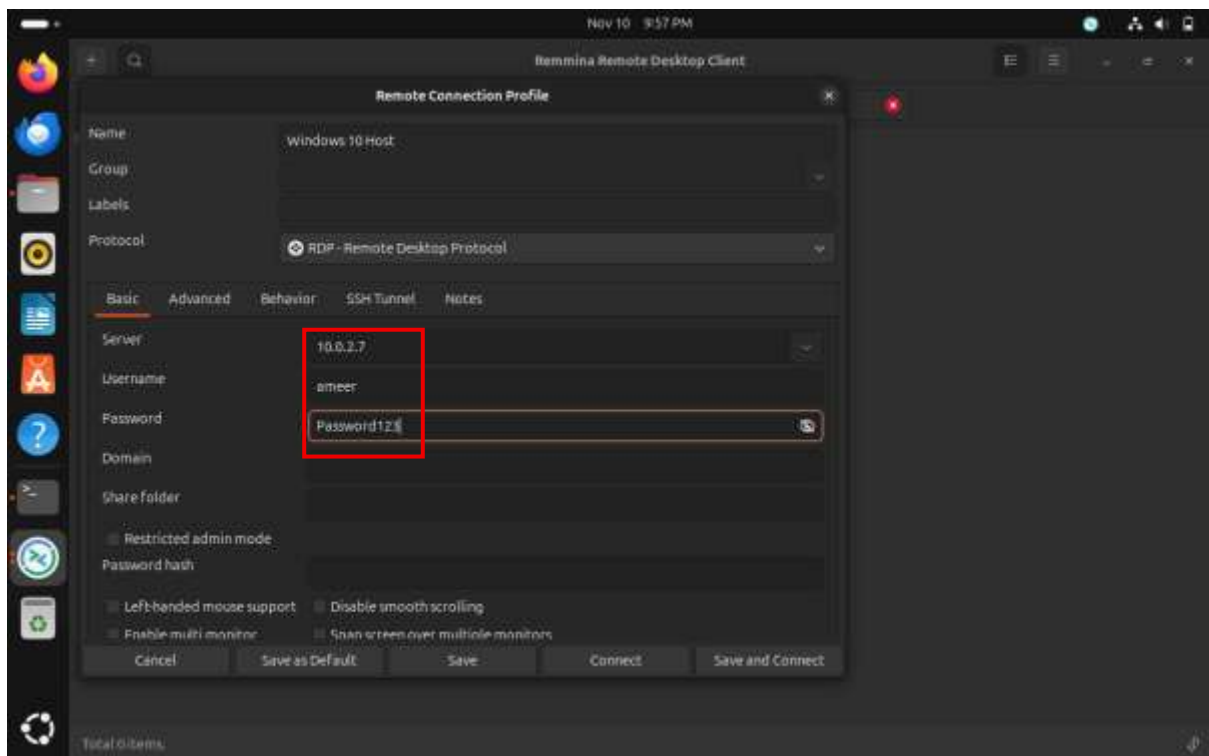
Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::3e43:82b4:5f3f:eb5c%5
    IPv4 Address. . . . . : 10.0.2.7
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 10.0.2.1
PS C:\Users\ameer>
```

User Name: Enter the Windows 10 username.

```
Ethernet adapter Ethernet:  
  
    Connection-specific DNS Suffix  . :  
    Link-local IPv6 Address . . . . . : fe80::3e43:82b4:5f3f:eb5c%5  
    IPv4 Address. . . . . : 10.0.2.7  
    Subnet Mask . . . . . : 255.255.255.0  
    Default Gateway . . . . . : 10.0.2.1  
PS C:\Users\ameer> whoami  
desktop-e9p950c\ameer  
PS C:\Users\ameer>
```

Fill the relevant fields:



Configure Drive Mapping (Crucial Step):

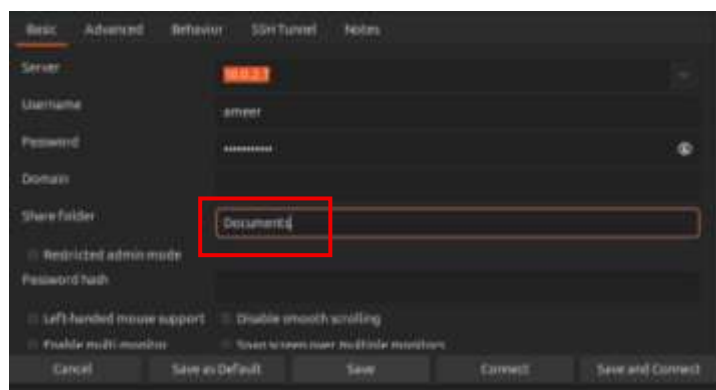
Find the share folder field within the Remmina profile editor.

Select the local Linux folder you want to map.

Turn off the windows firewall:

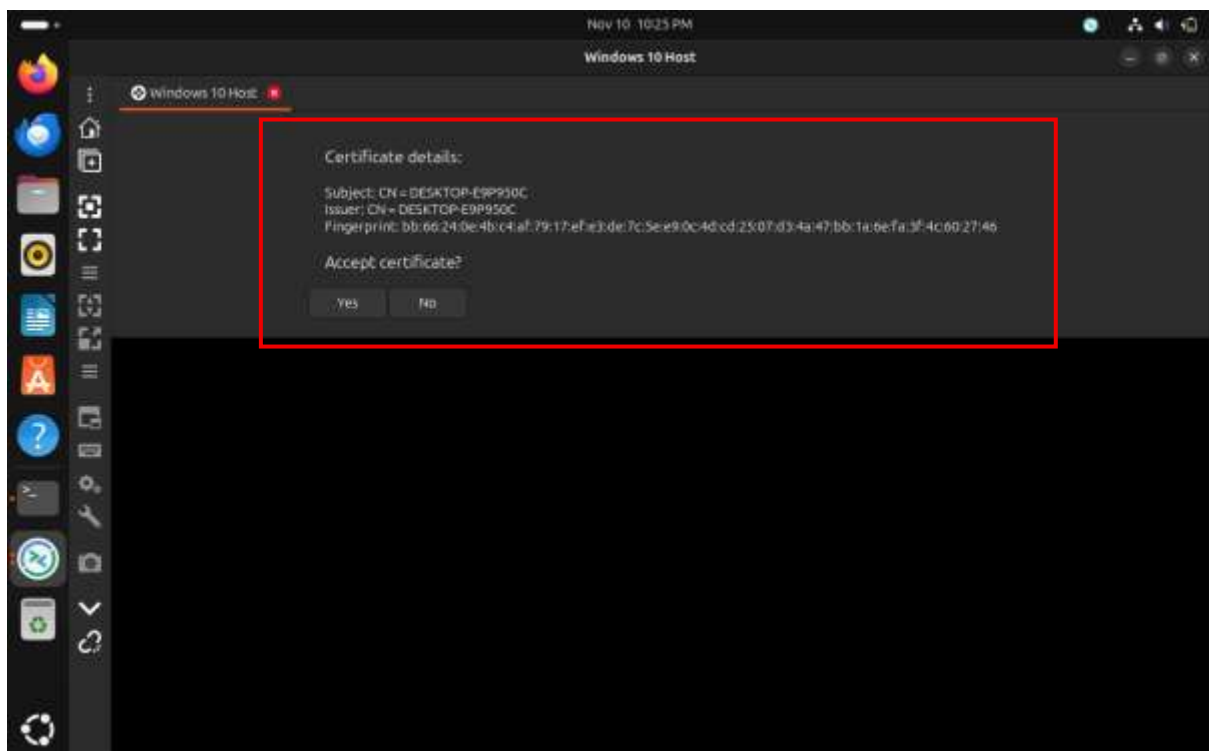


Save the profile and Connect.



Verification (Windows 10):

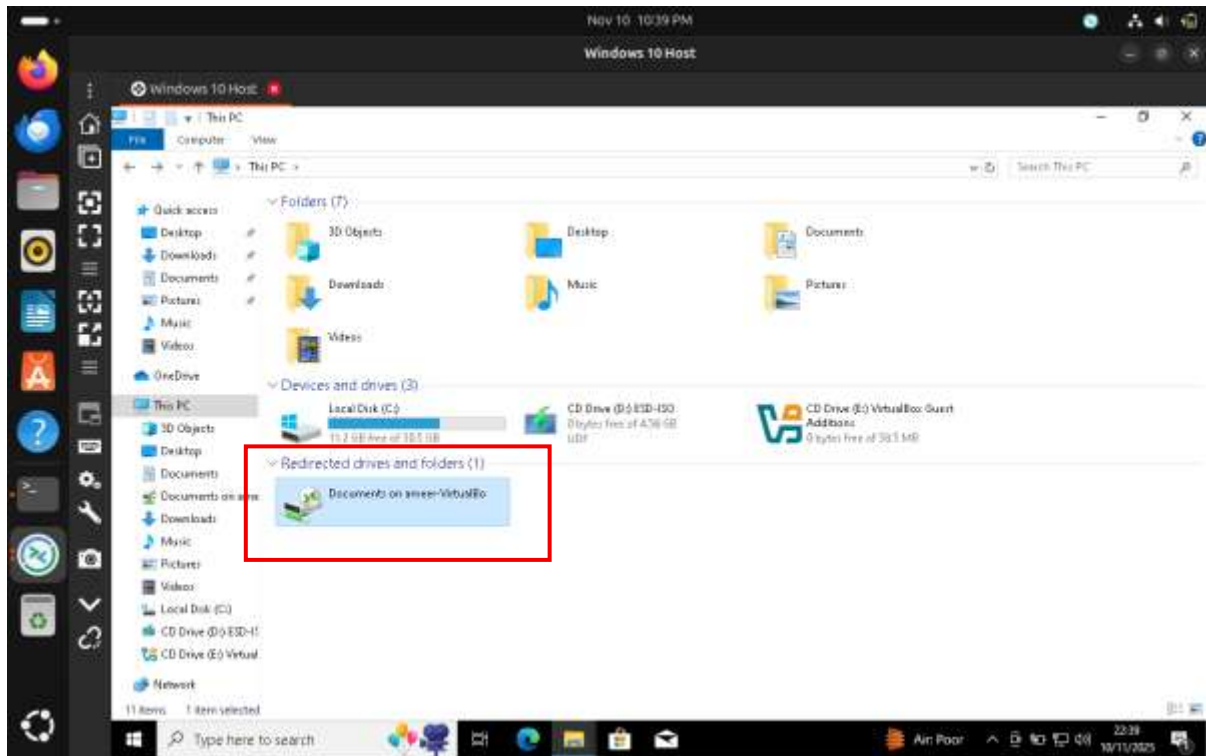
Check File Explorer: Once the RDP session is established from the Linux machine, on the remote Windows 10 desktop:



Open File Explorer on the Windows 10 desktop.

Go to "This PC".

Look under "Network locations" or "Devices and drives".



Question 3- Users and File Configurations

3.1 Add an additional user account for Frank Black

Command: Add the user fblack.

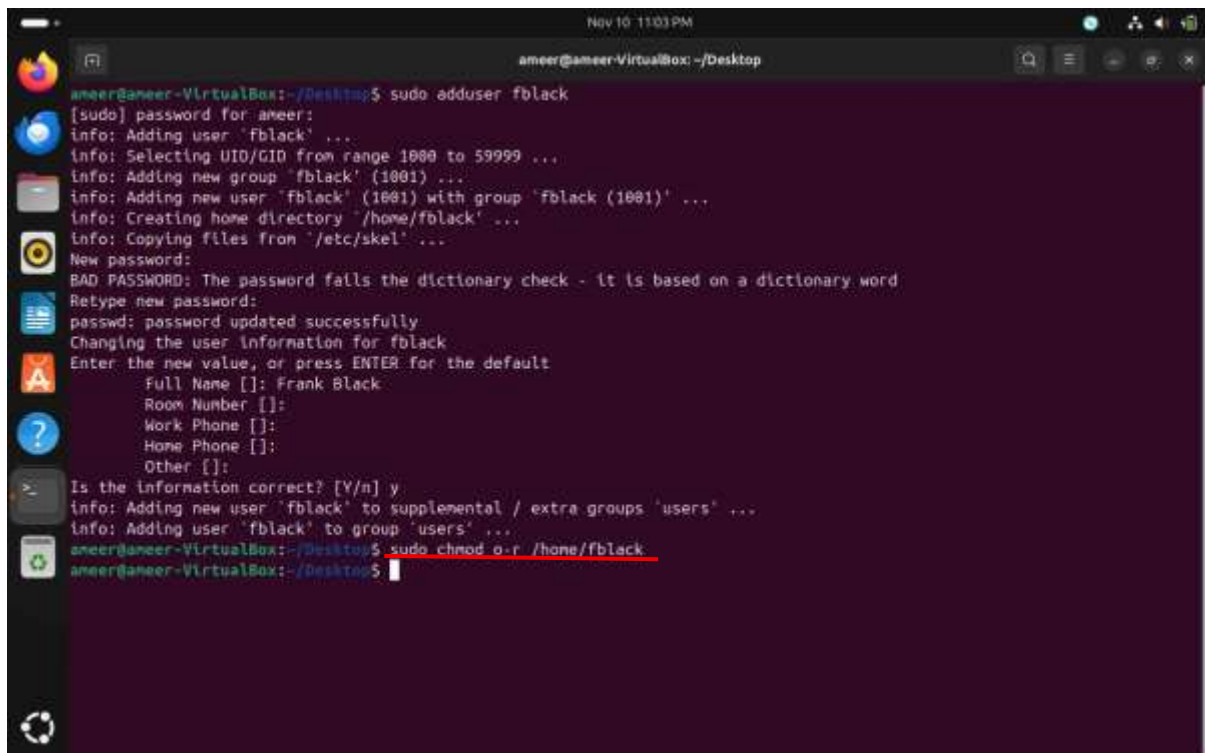
“sudo adduser fblack”

```
aneer@aneer-VirtualBox: ~/Desktop$ sudo adduser fblack
[sudo] password for aneer:
info: Adding user 'fblack' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group 'fblack' (1001) ...
info: Adding new user 'fblack' (1001) with group 'fblack (1001)' ...
info: Creating home directory '/home/fblack' ...
info: Copying files from '/etc/skel' ...
New password:
BAD PASSWORD: The password fails the dictionary check - it is based on a dictionary word
Retype new password:
passwd: password updated successfully
Changing the user information for fblack
Enter the new value, or press ENTER for the default
  Full Name []: Frank Black
    Room Number []:
```

3.2 Remove world readable permissions from Frank Black's home folder

Remove read permission for others (world) from the home folder.

“sudo chmod o-r /home/fblack”

A terminal window titled 'ameer@ameer-VirtualBox: ~/Desktop' showing the process of adding a user 'fblack'. The user is added with a group 'fblack' and a home directory '/home/fblack'. A password is set, and user information is entered. The command 'sudo chmod o-r /home/fblack' is executed and highlighted in red.

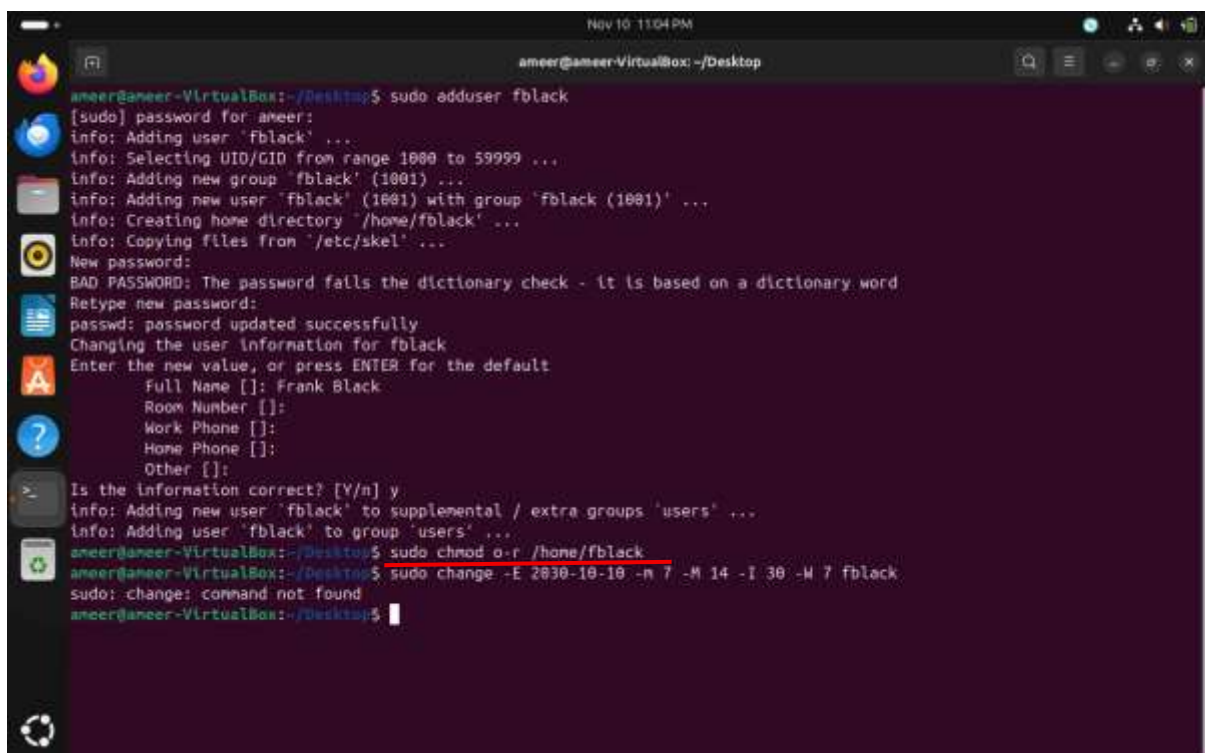
```
ameer@ameer-VirtualBox:~/Desktop$ sudo adduser fblack
[sudo] password for ameer:
info: Adding user 'fblack' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group 'fblack' (1001) ...
info: Adding new user 'fblack' (1001) with group 'fblack (1001)' ...
info: Creating home directory '/home/fblack' ...
info: Copying files from '/etc/skel' ...
New password:
BAD PASSWORD: The password falls the dictionary check - It is based on a dictionary word
Retype new password:
passwd: password updated successfully
Changing the user information for fblack
Enter the new value, or press ENTER for the default
    Full Name []: Frank Black
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
info: Adding new user 'fblack' to supplemental / extra groups 'users' ...
info: Adding user 'fblack' to group 'users' ...
ameer@ameer-VirtualBox:~/Desktop$ sudo chmod o-r /home/fblack
ameer@ameer-VirtualBox:~/Desktop$
```

(This changes drwxr-xr-x to drwxr-x--x.)

3.3 Apply account policy using a single command

Use the chage tool to apply all required policies at once.

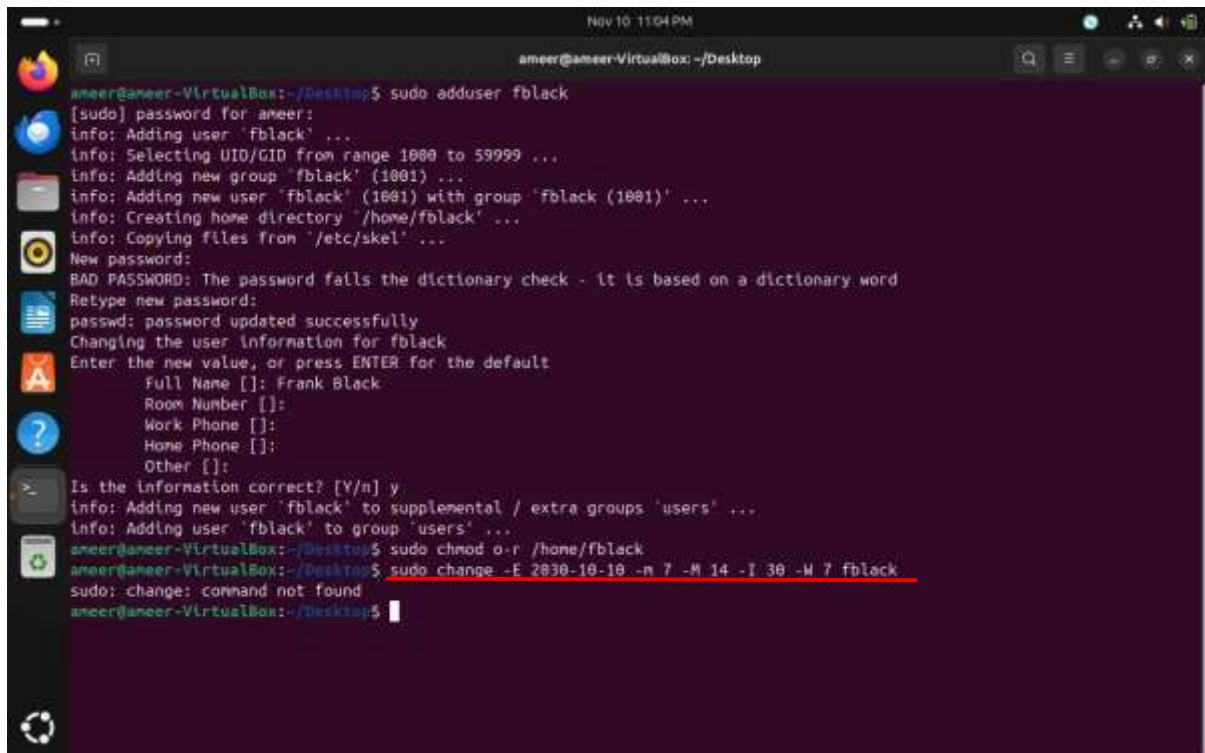
“sudo chage -E 2030-10-10 -m 7 -M 14 -I 30 -W 7 fblack”

A terminal window titled 'ameer@ameer-VirtualBox: ~/Desktop' showing the same user creation process as the previous screenshot. After the user is created, the command 'sudo chage -E 2030-10-10 -m 7 -M 14 -I 30 -W 7 fblack' is entered and highlighted in red. The terminal shows an error message 'sudo: chage: command not found' because the 'chage' command is not installed.

```
ameer@ameer-VirtualBox:~/Desktop$ sudo adduser fblack
[sudo] password for ameer:
info: Adding user 'fblack' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group 'fblack' (1001) ...
info: Adding new user 'fblack' (1001) with group 'fblack (1001)' ...
info: Creating home directory '/home/fblack' ...
info: Copying files from '/etc/skel' ...
New password:
BAD PASSWORD: The password falls the dictionary check - It is based on a dictionary word
Retype new password:
passwd: password updated successfully
Changing the user information for fblack
Enter the new value, or press ENTER for the default
    Full Name []: Frank Black
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
info: Adding new user 'fblack' to supplemental / extra groups 'users' ...
info: Adding user 'fblack' to group 'users' ...
ameer@ameer-VirtualBox:~/Desktop$ sudo chmod o-r /home/fblack
ameer@ameer-VirtualBox:~/Desktop$ sudo chage -E 2030-10-10 -m 7 -M 14 -I 30 -W 7 fblack
sudo: chage: command not found
ameer@ameer-VirtualBox:~/Desktop$
```

Flags:

- E 2030-10-10: Set account expiration.
- m 7: Set minimum password age to 7 days.
- M 14: Set maximum password age to 14 days.
- I 30: Set inactivity period to 30 days.
- W 7: Set warning time period to 7 days.

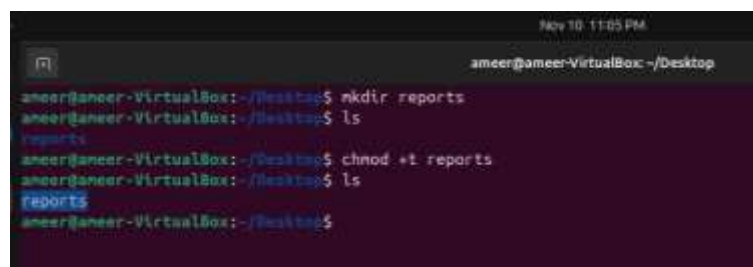


```
ameer@ameer-VirtualBox: ~/Desktop
ameer@ameer-VirtualBox:~/Desktop$ sudo adduser fblack
[sudo] password for ameer:
info: Adding user 'fblack' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group 'fblack' (1001) ...
info: Adding new user 'fblack' (1001) with group 'fblack (1001)' ...
info: Creating home directory '/home/fblack' ...
info: Copying files from '/etc/skel' ...
New password:
BAD PASSWORD: The password falls the dictionary check - It is based on a dictionary word
Retype new password:
passwd: password updated successfully
Changing the user information for fblack
Enter the new value, or press ENTER for the default
    Full Name []: Frank Black
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
info: Adding new user 'fblack' to supplemental / extra groups 'users' ...
info: Adding user 'fblack' to group 'users' ...
ameer@ameer-VirtualBox:~/Desktop$ sudo chmod o-r /home/fblack
ameer@ameer-VirtualBox:~/Desktop$ sudo change -E 2030-10-10 -m 7 -M 14 -I 30 -W 7 fblack
sudo: change: command not found
ameer@ameer-VirtualBox:~/Desktop$
```

3.4 Create directory and modify permissions

Create Directory: Create the reports directory in your home folder.

“mkdir ~/reports”



```
ameer@ameer-VirtualBox: ~/Desktop
ameer@ameer-VirtualBox:~/Desktop$ mkdir reports
ameer@ameer-VirtualBox:~/Desktop$ ls
reports
ameer@ameer-VirtualBox:~/Desktop$ chmod +t reports
ameer@ameer-VirtualBox:~/Desktop$ ls
reports
ameer@ameer-VirtualBox:~/Desktop$
```

Apply Sticky Bit: Modify permissions using the Sticky Bit (+t) to ensure users can only delete files they created.

“chmod +t ~/reports”

```

ameer@ameer-VirtualBox:~/Desktop$ mkdir reports
ameer@ameer-VirtualBox:~/Desktop$ ls
reports
ameer@ameer-VirtualBox:~/Desktop$ chmod +t reports
ameer@ameer-VirtualBox:~/Desktop$ ls
reports
ameer@ameer-VirtualBox:~/Desktop$

```

(Permissions will look like drwxrwxrwt or drwxrwxxt).

3.5 Determine the read performance of the device sda1

Install the hdparm utility.

“sudo apt install hdparm”

```

ameer@ameer-VirtualBox:~/Desktop$ sudo apt install hdparm
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
hdparm is already the newest version (9.65+ds-1build1).
hdparm set to manually installed.
The following packages were automatically installed and are no longer required:
  libgl1-amber-dri libglapi-mesa libllvm19
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 1 not upgraded.
ameer@ameer-VirtualBox:~/Desktop$ sudo hdparm -t /dev/sda1

/dev/sda1:
Timing buffered disk reads: read(2097152) returned 1048576 bytes
ameer@ameer-VirtualBox:~/Desktop$

```

Measure the buffered read speed of the /dev/sda1 partition.

“sudo hdparm -t /dev/sda1”

```

ameer@ameer-VirtualBox:~/Desktop$ sudo apt install hdparm
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
hdparm is already the newest version (9.65+ds-1build1).
hdparm set to manually installed.
The following packages were automatically installed and are no longer required:
  libgl1-amber-dri libglapi-mesa libllvm19
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 1 not upgraded.
ameer@ameer-VirtualBox:~/Desktop$ sudo hdparm -t /dev/sda1

/dev/sda1:
Timing buffered disk reads: read(2097152) returned 1048576 bytes
ameer@ameer-VirtualBox:~/Desktop$

```

Final speed test: read 92097152) returned 1048576 bytes

3.6 Change the current runlevel to 6

Use the traditional init command to change the runlevel to 6 (which initiates a Reboot).

“sudo init 6”

The Ubuntu Virtual Machine will immediately reboot.

Question 4 – Scripting

4.1 Create the update script file

Use nano to create the script file.

“nano update”

Add:

“sudo apt update && sudo apt upgrade -y”

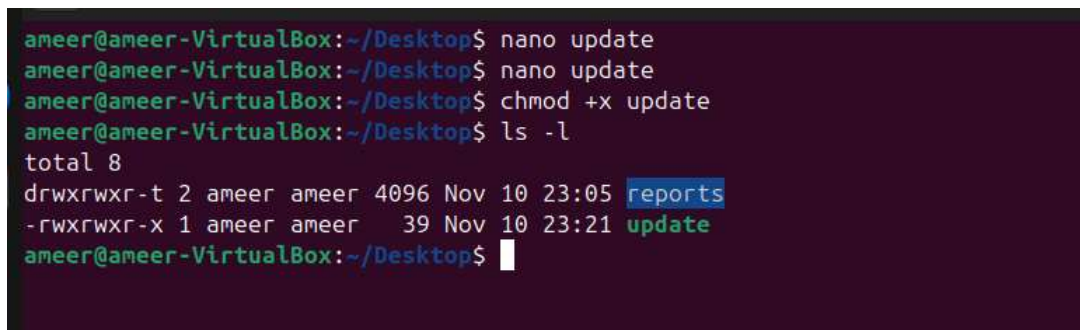
A screenshot of a terminal window showing the GNU nano 7.2 editor. The file being edited is named 'update'. The content of the file is the command 'sudo apt update && sudo apt upgrade -y'.

Save the file.

4.2 Assign executable permissions

Use chmod to add execute permission (+x).

“chmod +x update”

A screenshot of a terminal window showing a series of commands and their output. The commands are: 'nano update', 'nano update', 'chmod +x update', and 'ls -l'. The output of 'ls -l' shows two files: 'reports' and 'update'. The 'update' file has permissions '-rwxrwxr-x' and is owned by 'ameer'.

Verify using “ls -l update” should show an x in the permissions.

4.3 Create a new crontab entry

Open the user's crontab for editing.

“crontab -e”

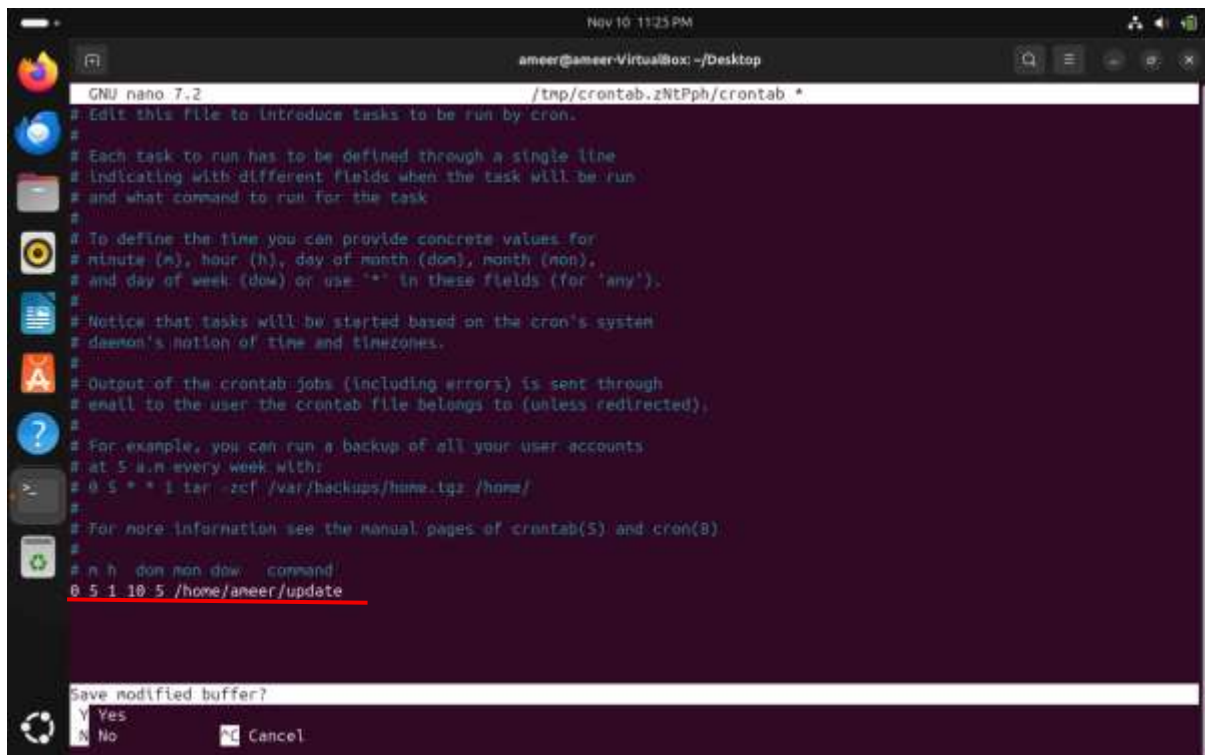
The schedule: 05:00 AM, Day of Month 1, Month 10, Day of Week 5 (Friday).

Fields: Minute Hour Day_of_Month Month Day_of_Week

Values: 0 5 1 10 5

Add the following line to the crontab file, using the full path to your script (/home/yourusername/update).

“0 5 1 10 5 /home/yourusername/update”



```
Nov 10 11:25 PM
ameer@ameer-VirtualBox: ~/Desktop
GNU nano 7.2 /tmp/crontab.zNtPph/crontab *
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use "*" in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow   command
0 5 1 10 5 /home/ameer/update

Save modified buffer?
Y Yes
N No
C Cancel
```

Save the crontab file (installs the new scheduled job).

References List

<https://www.google.com/url?sa=E&q=https%3A%2F%2Fwww.virtualbox.org%2Fmanual%2Fch06.html>

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